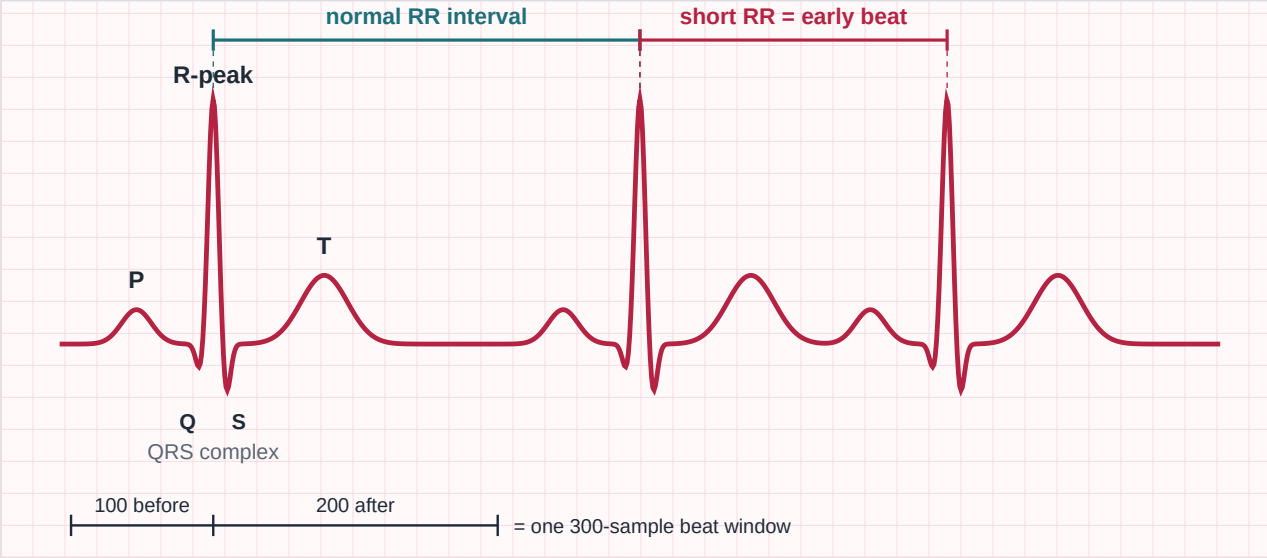
Project Glossary

SmartHealthcare-Agentic: Agentic Cardiac AI

Every important term, explained in plain language



A stylised ECG: three heartbeats. The third arrives early (a short RR interval), which is exactly the clue used to spot S (supraventricular) beats.

How to use this document. Terms are grouped by topic and ordered so each section builds on the ones before it. Every entry first explains the idea for someone with no medical or programming background, then (in *teal italics*) says what it means specifically in this project. The key numbers are collected on the last page.

Not a medical device. Everything in this project is research only.

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The whole project in one line

Raw heart recordings go in on the left. The teal steps are the Phase 1 data pipeline; the red steps are the Phase 1b modelling and evaluation.

In one sentence: the project downloads 48 expert-labelled heart recordings, cleans them, cuts them into individual heartbeats, sorts each beat into one of five types, splits the patients so the model is always tested on people it has never seen (DS1 for learning, DS2 for testing), and measures how well a classifier recognises each beat type.

1. The big picture

What the project is trying to do, in one page.

Smart Healthcare / Agentic Cardiac AI	The overall goal: software that reads heart recordings and patient data and helps flag heart problems. "Agentic" means the finished system would not just give a score but take reasoned steps on its own (e.g. decide what to check next), like a careful assistant. *In this project:* only the foundation is built so far: cleaning heart-signal data and testing simple classifiers. Nothing here is a medical device or is used on real patients.
Phase	A stage of the work. The project is split into phases so each part can be finished and checked before the next one starts. *In this project:* Phase 0 = set up computer and download data. Phase 1 = clean and prepare the heart data. Phase 1b = first simple classifier (the baseline). Phase 1c = results web page. Phase 1d = neural network design (planned). Phase 2 = MIMIC-III hospital data. Phase 3 = signal-quality layer. Phase 4 = combining everything into the agent.
Pipeline	A fixed series of steps that data passes through, like an assembly line: load, clean, cut into beats, label, save. Running it again on the same input gives the same output.
Preprocessing	Everything done to raw data before a model learns from it: removing noise, cutting it into pieces, putting numbers on the same scale, attaching labels.
Model / Classifier	A program that learns from examples and then sorts new items into categories. Here it looks at one heartbeat and says which type of beat it is (normal, early, ventricular, ...).
Baseline	A deliberately simple first model. It is the "floor": any fancier model must beat it to prove it learned something extra. *In this project:* a Random Forest using RR + shape features (Phase 1b).

2. Heart and ECG basics

The medical words, explained without medical training.

ECG (electrocardiogram)	A recording of the tiny electrical signals the heart makes each time it beats, picked up by stickers (electrodes) on the skin. It shows up as a wiggly line that repeats with every beat.
Heartbeat and its waves (P, QRS, T)	Each beat draws a familiar pattern. The small **P wave** is the upper chambers (atria) firing. The tall, sharp **QRS complex** is the main pumping chambers (ventricles) firing. The rounded **T wave** is the ventricles resetting. See the picture on the cover page.

R-peak	The highest, sharpest point of the QRS spike. It is the easiest point to find in each beat, so it is used as the "timestamp" of the beat. *In this project:* every saved beat is lined up so its R-peak sits at the same position (sample 100).
Arrhythmia	Any irregular heartbeat: too early, too late, the wrong shape, or starting in the wrong place. Detecting these is what the classifiers are for.
Lead (MLII, V5)	One "viewing angle" of the heart's electricity, set by which electrodes are compared. Different leads make the same beat look different. **MLII** (modified limb lead II) is the usual main lead in MIT-BIH; **V5** is a chest lead. *In this project:* the lead is chosen by its *name*, not by its column position. Records 102 and 104 have no MLII, so they fall back to V5 instead of silently mixing leads.
Sampling rate / Hz / sample	A digital recording measures the signal many times per second. Each measurement is a **sample**. **Hz** (hertz) means "per second". *In this project:* MIT-BIH is recorded at **360 Hz**: 360 samples every second. So 300 samples = 0.83 s.
Pacemaker / paced beat	A pacemaker is a small implanted device that makes the heart beat with an electrical pulse. A paced beat is one triggered by the device, not the heart itself. *In this project:* records **102, 104, 107, 217** come from patients with pacemakers. The standard (AAMI) rules exclude them from training and testing.
Premature beat / prematurity	A beat that arrives earlier than expected. Arriving early is the key sign of an S beat (supraventricular). It is about *timing*, not shape.
Compensatory pause	The longer-than-normal gap that often follows an early ventricular (V) beat, as the heart "catches up" to its normal rhythm.
Ventricular flutter wave (symbol !)	A very fast, wave-like electrical pattern in the ventricles during a flutter episode. It is not a separate, countable beat. *In this project:* its 472 marks (all in record 207) are excluded, not counted as V beats.

3. The data sources

Where the heart recordings come from and the file formats involved.

PhysioNet	A free, trusted online library of medical research recordings run by MIT. Both datasets in this project come from it.
MIT-BIH Arrhythmia Database	A famous collection of **48 half-hour, two-lead ECG recordings** from 47 people, made at Boston's Beth Israel Hospital (BIH) and MIT. Cardiologists marked every single beat by hand, so it is the standard test set for heartbeat classifiers. *In this project:* version 1.0.0, 93.9 MB, openly licensed (no account needed). It has 112,647 annotations and 109,494 beats.

Record	One recording from one patient session, named by a number like 100, 101, ... 234. Because each record is one person, *record* and *patient* are used almost interchangeably here.
Annotation	A label placed at a moment in the recording by an expert, e.g. "normal beat here" or "rhythm changes here". Each has a one-character **symbol** (N, V, A, +, ~ ...).
Beat annotation vs. non-beat annotation	A beat annotation marks an actual heartbeat and its type. A non-beat annotation marks something else: a rhythm change (+), signal quality change (~), a comment ("), etc. *In this project:* the 3,153 non-beat annotations are **excluded** and counted, never turned into fake beats.
WFDB format (.dat / .hea / .atr)	PhysioNet's file format. Each record is three files: **.dat** = the raw signal numbers, **.hea** = a header describing it (rate, leads), **.atr** = the expert annotations. **wfdb** is also the name of the Python library that reads them.
MIMIC-III	A large database of real hospital intensive-care (ICU) records: vital signs, lab tests, medications, notes. Planned for Phase 2, not used yet. *In this project:* three versions exist: a small open **Demo** (~100 patients), the **full** database (needs credentialing), and a **Waveform** subset (terabytes, too large for this machine).
Credentialed access / CITI / DUA	To use the full MIMIC-III you must prove you are a trained researcher: make a PhysioNet account, complete **CITI** human-subjects research training, and sign a **Data Use Agreement (DUA)** promising to protect the data. Only the person can do this, and it takes days to weeks.
De-identified data	Patient data with names and identifying details removed. It is still private medical data and is kept only on the local machine, never uploaded or committed to git.

4. Beat classes (the labels)

The categories every heartbeat is sorted into.

AAMI	The Association for the Advancement of Medical Instrumentation. It publishes the standard rules for testing heartbeat-detection devices, including which 5 classes to use and which records to leave out. Following it makes results comparable with other research.

AAMI 5 classes: N, S, V, F, Q	The dozens of MIT-BIH beat symbols are grouped into five classes, always kept in this order: **N - Normal**: ordinary beats (also includes bundle branch block and escape beats). **S - Supraventricular**: early beats starting *above* the ventricles (e.g. atrial premature beats). Shape often looks normal; the clue is that it comes early. **V - Ventricular**: beats starting in the ventricles (e.g. PVCs). They look wide and odd. **F - Fusion**: a mix of a normal and a ventricular beat happening at once. **Q - Unclassifiable / paced**: beats that do not fit, or were triggered by a pacemaker.
PVC (premature ventricular contraction)	The most common V beat: an early beat started by the ventricles, with a wide, unusual shape.
EXCLUDED / unknown symbol	EXCLUDED = a known non-beat annotation that is set aside and counted. Unknown = a symbol not in the mapping table at all; in strict mode the program stops with an error so gaps are never hidden. *In this project:* 0 unknown symbols were found in the real database.
Why Q is "not a result"	Almost all Q beats are paced beats in the 4 excluded records. After following AAMI rules, only 13 Q beats remain for training and 7 for testing. That is far too few to learn or to measure, so Q scores are reported but not trusted.

5. Cleaning the signal

How noise is removed without damaging the heartbeat.

Noise / artifact	Anything in the recording that is not the heart: body movement, breathing, muscle twitches, electrical hum from the building.
Baseline wander	A slow up-and-down drift of the whole signal, mostly caused by breathing and movement. It is very slow (well under 0.5 times per second).
Powerline interference (50/60 Hz)	A steady hum from mains electricity: 60 Hz in the US, 50 Hz in Europe.
Filter	A mathematical tool that removes some "speeds" (frequencies) of wiggle from a signal and keeps others, like an audio equalizer that cuts the bass and treble.
Bandpass filter (0.5 - 40 Hz)	A filter that keeps only a middle band of frequencies. Below 0.5 Hz removes baseline wander; above 40 Hz removes muscle noise and powerline hum. The heartbeat's shape lives inside that band.
Butterworth filter	A classic filter design that keeps the kept band very smooth and flat. "Order 3" describes how sharply it cuts off at the edges.

Notch filter	A filter that removes one narrow frequency, e.g. exactly 60 Hz. *In this project:* built but **off by default**: the 40 Hz bandpass already removes 60 Hz hum, so the notch would do nothing. Flag: `--notch`.
Zero-phase filtering (filtfilt vs. lfilter)	An ordinary filter (**lfilter**) slightly delays the signal, shifting every R-peak away from where the expert marked it. **filtfilt** runs the filter forwards and then backwards, so the delays cancel out and peaks stay exactly in place. This keeps every beat correctly aligned.
NaN / inf	"Not a Number" / infinity: broken values in the data. One NaN spreads through a filter and ruins a whole recording, so the pipeline rejects them loudly instead of continuing.

6. Cutting and scaling beats

Turning a 30-minute recording into thousands of equal-sized beat snapshots.

Segmentation	Cutting a long recording into smaller pieces. **Beat-level** segmentation cuts one piece per heartbeat; **window-level** cuts fixed 5 or 10-second chunks (for later rhythm models).
Beat window (300 samples: 100 before, 200 after)	Each beat is cut from 100 samples before its R-peak to 200 samples after it: 300 samples, about 0.83 seconds. This captures the P wave, QRS and T wave of that beat.
Boundary beat / zero-padding	A beat too close to the very start or end of a recording has no full window around it. Filling the missing part with zeros (zero-padding) would invent a fake flat line, so such beats are **dropped and counted** instead. *In this project:* 48 beats were dropped this way.
Normalization / z-score	Putting numbers on a common scale. A z-score subtracts the average and divides by the spread (standard deviation), so values end up centred on 0 with a typical size of 1. This stops differences in recording volume between patients from confusing the model.
Fit / apply (train-only statistics)	"Fit" = calculate the average and spread. "Apply" = use them to rescale data. They are **fitted on training records only** and then applied to all splits, so the test data never influences the preparation (that would be leakage).
Epsilon / flatline guard	A tiny number added when dividing, so a perfectly flat signal (spread of 0) gives zeros instead of a divide-by-zero error.

7. Splitting the data fairly (DS1 / DS2)

The most important idea for trusting any result in this project.

Train / validation / test split	Data is divided into three groups. **Training** = examples the model learns from (the textbook). **Validation** = used to tune settings and choose between models (practice exams). **Test** = kept locked away and used once at the end to report the score (the final exam). If the final exam questions leak into the textbook, the score means nothing.
Data leakage	When information from the test data sneaks into training or preparation. The model looks brilliant on paper but fails on new patients. The project is built around preventing it.
Record-wise split vs. beat-level split	A **beat-level** split scatters beats randomly, so beats from the *same patient* land in both training and test. The model just recognises that patient's heart and scores unrealistically well. A **record-wise** split keeps every patient entirely on one side. *In this project:* beat-level splitting is not even possible in the code, by design.
Inter-patient evaluation	Testing only on patients the model has never seen. This is the honest, real-world test: a new patient walks in and the model has never met them.
DS1 and DS2	The standard inter-patient split of MIT-BIH, introduced by **de Chazal et al. (2004)**. The 44 non-paced records are divided into two equal groups of 22: **DS1** (dataset 1, *training*): 101, 106, 108, 109, 112, 114, 115, 116, 118, 119, 122, 124, 201, 203, 205, 207, 208, 209, 215, 220, 223, 230. **DS2** (dataset 2, *testing*): 100, 103, 105, 111, 113, 117, 121, 123, 200, 202, 210, 212, 213, 214, 219, 221, 222, 228, 231, 232, 233, 234. Everyone in the field uses the same lists, so results from different papers can be compared. *In this project:* the lists were checked against two published sources before being used. A few DS1 records are set aside as the validation group, so the model trains on 18 records and is tested on all 22 DS2 records (about 49,700 beats). Use flag `--strategy ds1_ds2`.
de Chazal et al. (2004)	The research paper (IEEE Transactions on Biomedical Engineering) that defined DS1/DS2 and reported classic results: V sensitivity 0.803, S sensitivity 0.760. This project uses their record lists only; its method differs (one lead instead of two, different features and classifier), so the numbers are a reference range, not a like-for-like contest.
Seeded record-wise split (31 / 7 / 6)	The default alternative split: records are shuffled randomly and divided about 70/15/15 into 31 training, 7 validation and 6 test records. A **seed** is a fixed starting number for the randomness, so the "random" shuffle is identical every run.

"DS2 is partially spent"	Every look at the test set risks nudging decisions toward it. DS2 was evaluated three times in Phase 1b (for debugging), so it is treated as slightly used; future tuning must use validation records only, and DS2 is touched once for final reporting.
Leakage tripwire	A safety check in the training script. If the data was prepared (normalized) for one split but a model is trained on a different split, it refuses to run and prints the command that fixes it.

8. Features: what the model looks at

The measurements taken from each beat. RR = timing, morphology = shape.

Feature	One measurable number describing an item, like height or weight for a person. A model makes its decision from a list of features.
RR interval	The time between the R-peak of one beat and the R-peak of the next, in seconds. It is simply the gap between heartbeats. A resting heart at 70 beats per minute has RR of about 0.86 s.
pre_rr / post_rr	**pre_rr** = the gap from the previous beat to this beat. **post_rr** = the gap from this beat to the next one. An early (S) beat has a short pre_rr; a V beat is often followed by a long post_rr (the compensatory pause).
local_rr	The average gap over the nearby beats (about 5 on each side). It describes the patient's rhythm right at that moment.
global_rr	The average gap over the whole recording of that patient. It describes that person's usual heart rate.
pre_rr_ratio / post_rr_ratio	pre_rr (or post_rr) divided by global_rr. Dividing by the patient's own average makes "early" mean the same for a fast-hearted and a slow-hearted person. A ratio of 0.6 means "arrived at 60% of this person's normal gap". *In this project:* a single rule, pre_rr_ratio < 0.85, catches **97.2%** of S beats in DS2, showing the timing clue is strong.
Morphology	The *shape* of the beat's waveform: how tall, wide and pointed the waves are.
Mean pooling (300 to 30)	Shrinking the 300-sample beat by averaging every 10 samples into one number, giving 30 shape features. Averaging (instead of just taking every 10th sample) keeps the sharp QRS spike from being skipped over.
Feature set: rr+morphology / rr_only / morphology_only	The three feature combinations tested: all 36 features (6 RR + 30 shape), timing only, or shape only. Comparing them shows which kind of information helps which class.

Ablation	An experiment that removes one part to see how much it mattered, like removing an ingredient from a recipe to see what it contributed.
Feature dilution	When a few very useful features are outnumbered by many less useful ones, so the model rarely gets to use the good ones. *In this project:* the 6 strong RR features were drowned out by 30 shape features. This is the diagnosed cause of the low S score, not class imbalance alone.

9. The classical models

The Phase 1b baseline classifiers and their settings.

Decision tree	A model that asks a series of yes/no questions ("is pre_rr_ratio below 0.85?") and follows the branches to an answer, like a flowchart.
Random Forest	Many decision trees, each built on slightly different random data and features, that vote on the answer. The crowd is more reliable than any single tree. *In this project:* the best baseline model.
max_features = 'sqrt'	A Random Forest setting: at each question, a tree may only consider a random handful of features (the square root of the total; about 6 of 36). This is why the 6 RR features were often not available at the key moment.
min_samples_leaf (2 vs. 50)	The minimum number of training beats that must end up at each final answer of a tree. With 2, trees grew huge and memorised individual patients; with 50, they had to learn general patterns. *In this project:* changing 2 to 50 raised S recall on DS2 from 0.063 to 0.271.
Overfitting / memorisation	When a model learns the specific training examples (e.g. one patient's unique beat shape) instead of the general rule, so it fails on new patients. Like memorising exam answers instead of the subject.
LDA (Linear Discriminant Analysis)	A simple, older classifier that separates classes by drawing straight-line boundaries between them. The method de Chazal et al. used.
Hyperparameter	A setting chosen by people before training (like min_samples_leaf or learning rate), as opposed to what the model learns by itself. Hyperparameters must be chosen on validation data, never test data.
scikit-learn (sklearn) Pipeline / scaler	scikit-learn is the Python machine-learning library used for the baseline. A **Pipeline** chains steps (e.g. a **scaler** that rescales features, then the classifier) so the scaler can only ever learn from training data.

10. Class imbalance

Why rare heartbeat types are hard, and the tools for handling it.

Class imbalance	When some categories have far more examples than others. In DS1 training data, for every S beat there are 26 normal beats; for every F beat, 79; for Q, over 4,800. A lazy model can just say "normal" every time and look accurate.
Class weights / inverse-frequency weights	Telling the model that mistakes on rare classes cost more. Weights are set inversely to how common a class is (rare = heavier). **class_weight="balanced"** is the scikit-learn option that does this automatically.
Weight cap	An upper limit on class weights. Q's raw weight would be about 1,125, which would let 13 beats overwhelm training; the cap prevents that (planned, Phase 1d).
Oversampling / undersampling	Oversampling = copying rare examples so they appear more often. Undersampling = throwing away some common examples. Both change the true mix of the data. *In this project:* not done during preprocessing, so the real class distribution is preserved for honest evaluation.
SMOTE	Synthetic Minority Over-sampling Technique: invents new rare examples by blending between existing ones. *In this project:* deliberately **not used**: blending beats from different patients creates heartbeats no real heart produced, and it has inflated published MIT-BIH results.
Focal loss	A training penalty that pays less attention to easy examples (obvious normal beats) and more to hard ones. **gamma** controls how strongly; gamma=0 makes it ordinary cross-entropy.
Threshold tuning	A model outputs a confidence for each class. Instead of always picking the highest, you can lower the bar for a rare class (e.g. call it S if S confidence is above 0.3). The cut-offs are tuned on validation data, with no retraining needed.

11. Measuring results (metrics)

How to read the numbers in the results tables.

Support	How many real examples of a class are in the test set. S has support 1,836 in DS2; Q only 7.
Confusion matrix	A table of real class (rows) versus predicted class (columns). The diagonal shows correct answers; everything else shows exactly which classes get mixed up (e.g. 1,290 real S beats called N).

True / false positive, false negative	Take class V. **True positive**: a real V called V. **False positive**: something else wrongly called V (a false alarm). **False negative**: a real V that was missed.
Recall = Sensitivity (Se)	Of all the real beats of a class, what fraction did the model find? "How many did we catch?" V recall 0.844 = 84.4% of ventricular beats found.
Precision = Positive Predictive Value (PPV)	Of all the beats the model *called* a class, what fraction were right? "How trustworthy is an alarm?" V precision 0.594 = about 3 of every 5 V alarms are real.
F1 score	A single number combining recall and precision (their harmonic mean). It is only high when *both* are high, so a model cannot cheat by shouting a class everywhere.
Macro average (macro recall, macro F1)	Calculate the score separately for each class, then take a simple average, so every class counts equally no matter how rare. **Macro recall** can be fooled: a model that labels thousands of normal beats as rare classes gets high recall for them. *In this project:* macro-F1 is the headline ranking metric from Phase 1d on. Best baseline: macro recall 0.459, macro F1 0.405.
Accuracy (and why it misleads)	The overall fraction of beats labelled correctly. Because 89% of DS2 beats are normal, a model that always says "normal" scores about 0.89 accuracy while finding zero arrhythmias. Reported but not trusted.
Reference range vs. reproduction	Landing in the same range as a published result shows the data carries the right signal. It is not a reproduction (same method) and does not prove a better method.

12. Neural network terms (Phase 1d, planned)

The design for the next model. Approved on paper, not built yet.

Neural network	A model made of many layers of simple connected units that together learn complex patterns directly from data, loosely inspired by brain neurons.
CNN (convolutional neural network) / Conv1d	A network that slides small pattern detectors along the signal to spot shapes (a sharp spike, a wide bump) wherever they occur. **Conv1d** is the version for 1-dimensional signals like ECG. **k** (kernel size) is how many samples each detector looks at at once.
Hybrid / two-branch model	The planned design: one branch (a CNN) reads the beat's **shape**; a second branch reads the 6 **RR timing** features. Their outputs are joined (**concatenated**) before the final decision, so timing cannot be drowned out by shape.
Why not a waveform-only CNN	Each beat is cut and centred on its R-peak, which erases when it arrived. An early beat and an on-time beat can look identical, so a shape-only model is blind to the main clue for S beats.

Layer vocabulary	**BatchNorm (BN)**: keeps numbers inside the network at a steady scale. **ReLU**: a simple on/off-style step that lets the network learn non-straight patterns. **MaxPool**: shrinks the signal by keeping the strongest value in each small region. **Global average pool**: summarises a whole signal into one number per detector. **Linear** layer: a basic fully-connected layer. **Dropout (0.3)**: randomly switches off 30% of units while training so the network cannot rely on any one path (reduces memorisation). **Parameters**: the ~100,000 numbers the network learns.
Cross-entropy loss	The standard score of how wrong a classifier's confidences are. Training tries to make it smaller. **Class-weighted** cross-entropy counts rare-class mistakes more.
Training vocabulary	**Epoch**: one full pass through all training data (max 50). **Batch**: the number of beats looked at before each update (256). **Adam**: a popular method for adjusting the network after each batch. **Learning rate** (0.001): how big each adjustment step is.
Early stopping / checkpoint	Keep checking validation score after each epoch; stop when it stops improving and keep the saved copy (**checkpoint**) from the best epoch. Prevents training too long and overfitting.
PyTorch / CPU wheel	PyTorch is the neural-network library planned for Phase 1d. A **wheel** is a ready-to-install package; the CPU version needs no graphics card, which is enough for this small dataset.
Arm / row (arms 0 - 5)	One variant in the planned experiment. All arms are identical except for one change: 0 = no imbalance handling, 1 = class weights, 2 = focal loss, 3 = + threshold tuning, 4 = oversampling, 5 = waveform-only (no RR branch). Changing one thing at a time shows *what caused* any improvement.
BiLSTM / sequence model	A network that reads several beats in a row, remembering what came before and after. Considered for later, out of scope for Phase 1d.

13. Software, testing and project files

Engineering words that appear in the repository and its documents.

Python / venv / requirements.txt	Python is the programming language. **venv** is a private, isolated set of installed packages for this project (Python 3.13.9). **requirements.txt** lists the exact package versions, so the setup can be recreated identically.
SMART_HEALTH_DATA / ~/physionet-data	An environment setting that tells the code where the data lives. By default `~/physionet-data`, deliberately outside the project folder so large private data never enters git.
.npz file (X, y, r_peaks)	A compressed NumPy file holding arrays. One per record: **X** = the beats (n beats x 300 samples), **y** = each beat's class number (0=N, 1=S, 2=V, 3=F, 4=Q), **r_peaks** = where each beat's R-peak is in the recording.

manifest.csv	A per-record bookkeeping table: lead used, sampling rate, beats kept, dropped, excluded, per-class counts. Its numbers must add up: kept + dropped + excluded + unknown = total annotations.
splits.json / normalization.json	**splits.json** records which records are in train, validation and test (and which were excluded). **normalization.json** stores the z-score averages from training records only, the records they came from, class counts and suggested class weights.
baseline_metrics.json / results/ folder	The training script writes its scores to baseline_metrics.json (overwritten each run). The **results/** folder keeps frozen, dated copies with written analysis, so later models can be compared against exact past numbers.
Test / pytest / fixture / synthetic data	A **test** is code that automatically checks other code works. **pytest** runs them (181 offline tests + 9 real-data tests). A **fixture** is prepared test input; here it is **synthetic** (fake but realistic) ECG, so tests need no download.
requires_data marker	A tag on the 9 tests that need the real MIT-BIH download. They are skipped by default.
Regression test	A test written after fixing a bug, to make sure the bug never comes back. It must be shown to *fail* on the buggy version, or it proves nothing.
Deterministic / reproducible / idempotent	**Deterministic**: same input always gives the same output. **Reproducible**: someone else can rerun it and get the same numbers. **Idempotent**: running it twice is the same as once (e.g. the download skips files already present).
Integrity check	Verifying a downloaded file is complete and readable (right sampling rate, not truncated).
Deviation / Problem Discovered	Terms used in IMPLEMENTATION.md. A **deviation** is a place where the work differed from the approved plan, written down with the reason. A **problem discovered** is a bug found and how it was fixed.
Provenance	The record of exactly how a result was produced: date, code version (commit), commands, package versions, settings.
Git / commit / GitHub Pages	**Git** tracks every version of the code; a **commit** is a saved snapshot. **GitHub Pages** hosts the project's generated results web page from the docs/ folder.

14. Key numbers at a glance

The figures most often quoted in the project documents.

What	Value
MIT-BIH records	48 (47 patients), 30 minutes each, 2 leads
Sampling rate	360 samples per second (360 Hz)
Total annotations	112,647
Beat annotations	109,494 (109,446 kept + 48 dropped at edges)
Non-beat annotations excluded	3,153
Paced records excluded	102, 104, 107, 217
Beat window	300 samples = 100 before + 200 after the R-peak (~0.83 s)
Bandpass filter	0.5 - 40 Hz, Butterworth, order 3, zero-phase
DS1 / DS2	22 training records / 22 test records (de Chazal 2004)
DS2 test beats	about 49,700
Features	6 RR (timing) + 30 morphology (shape) = 36
Best baseline	Random Forest, RR + morphology
Best baseline scores	macro recall 0.459, macro F1 0.405, accuracy 0.872
V (ventricular) recall	0.844 (published reference 0.803)
S (supraventricular) recall	0.271 (published reference 0.760)
S caught by one timing rule	0.972 using pre_rr_ratio < 0.85
Tests	181 offline + 9 real-data, all passing

Sources inside the repository: README.md, PlanDoc/PLAN.md, PlanDoc/IMPLEMENTATION.md, PlanDoc/specs/2026-09-16-phase-1d-hybrid-nn-design.md, results/2026-08-27-baseline-ds1_ds2.md, src/ecg/features.py and src/ecg/labels.py.